

Alexander Assal

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EDUCATION

California State Polytechnic University, Pomona

Bachelor of Science in Computer Engineering | Expected Spring 2027 | Upper Division Major GPA: 3.87

EXPERIENCE

Research Assistant — Autonomous Vehicle Laboratory

California State Polytechnic University, Pomona | 3801 W Temple Ave, Pomona, CA 91768 | Apr 2025 – Present

- Develop and deploy real-time computer vision systems for the AV2 autonomous vehicle platform.
- Design multi-model perception pipelines integrating YOLOv2 and YOLOv8 for simultaneous road segmentation, lane detection, and object recognition.
- Implement semantic segmentation algorithms achieving 93.2% mIoU for drivable area detection on the BDD100K benchmark.
- Collaborate with a cross-functional research team on autonomous vehicle navigation and safety systems.

Computer Vision Mentor — Autonomous Vehicle Laboratory Summer Mentorship

California State Polytechnic University, Pomona | Jun 2026 – Aug 2026

- Led the Computer Vision track of the AVL summer mentorship program, setting the technical roadmap and assigning mentee ownership across line detection, BEV geometry, ML obstacle detection, and ROS 2 integration.
- Replaced model-only lane detection with a classical CV baseline robust to shade and irregular markings.
- Authored the program's technical reference, onboarding worksheets, and biweekly reporting structure.

TECHNICAL PROJECTS

SteveROS Humanoid Robot Platform (Python, C++, ROS2) | Jan 2026 – Present

- Lead computer vision development for a 20-DOF humanoid robot platform, designing perception systems including vision-based pose estimation and real-time teleoperation retargeting.
- Implement MediaPipe pose tracking pipelines and VR controller integration (Meta Quest 3) for intuitive humanoid control and imitation learning data collection.
- Assist in developing the full ROS2 control stack, including MoveIt2 motion planning, custom CAN bus hardware interfaces, and MuJoCo physics simulation.
- Contribute to hardware bring-up, motor calibration, and multi-layer safety system implementation across the 20-DOF platform using Robstride RS-series actuators.

Multi-Modal Road Segmentation System (Python, PyTorch, OpenCV) | Apr 2025 – Present

- Built a real-time panoptic perception system for autonomous vehicles that detects drivable road areas, lane lines, vehicles, pedestrians, bicycles, traffic lights, and traffic cones.
- Integrated YOLOv2 (38.9M parameters) and YOLOv8m-seg models into a unified inference pipeline achieving 20–30 FPS on GPU with CUDA acceleration.
- Developed a desktop GUI application (1,600+ lines) using Tkinter with real-time FPS monitoring, adjustable confidence thresholds, and multi-source camera input support.
- Engineered a lighting-invariant traffic cone detection algorithm using HSV color-space analysis, geometric filtering, and morphological refinement.
- Created a Flask-based web interface for remote monitoring and a batch-processing segmentation editor for offline analysis.

TECHNICAL SKILLS

- Programming:** C, C++, Python, Java, HTML/XML
- Frameworks & Libraries:** PyTorch, OpenCV, NumPy, ROS/ROS2, MoveIt2, MuJoCo, MediaPipe, TensorFlow, Flask, Ultralytics (YOLOv8)
- Tools & Platforms:** MATLAB, Git/GitHub, Linux/Unix, Docker, Microsoft Excel, Microsoft Office
- Domains:** Semantic Segmentation, Object Detection, Deep Learning, Computer Vision, Humanoid Robotics, Motion Planning, CUDA/GPU Acceleration, Real-Time Systems